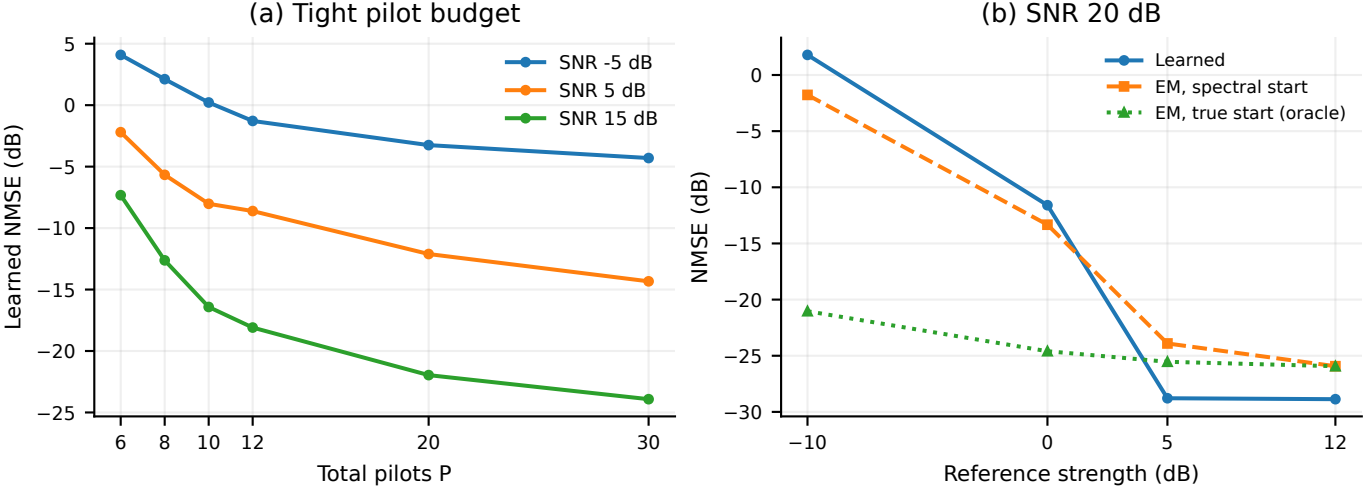
(a) Tight pilot budget
Learned NMSE (dB)
5
0
−5
−10
−15
−20
−25
6
8
10
12
20
30
Total pilots P
SNR -5 dB
SNR 5 dB
SNR 15 dB
(b) SNR 20 dB
NMSE (dB)
0
−5
−10
−15
−20
−25
−30
−10
0
5
12
Reference strength (dB)
Learned
EM, spectral start
EM, true start (oracle)